Exp no:11

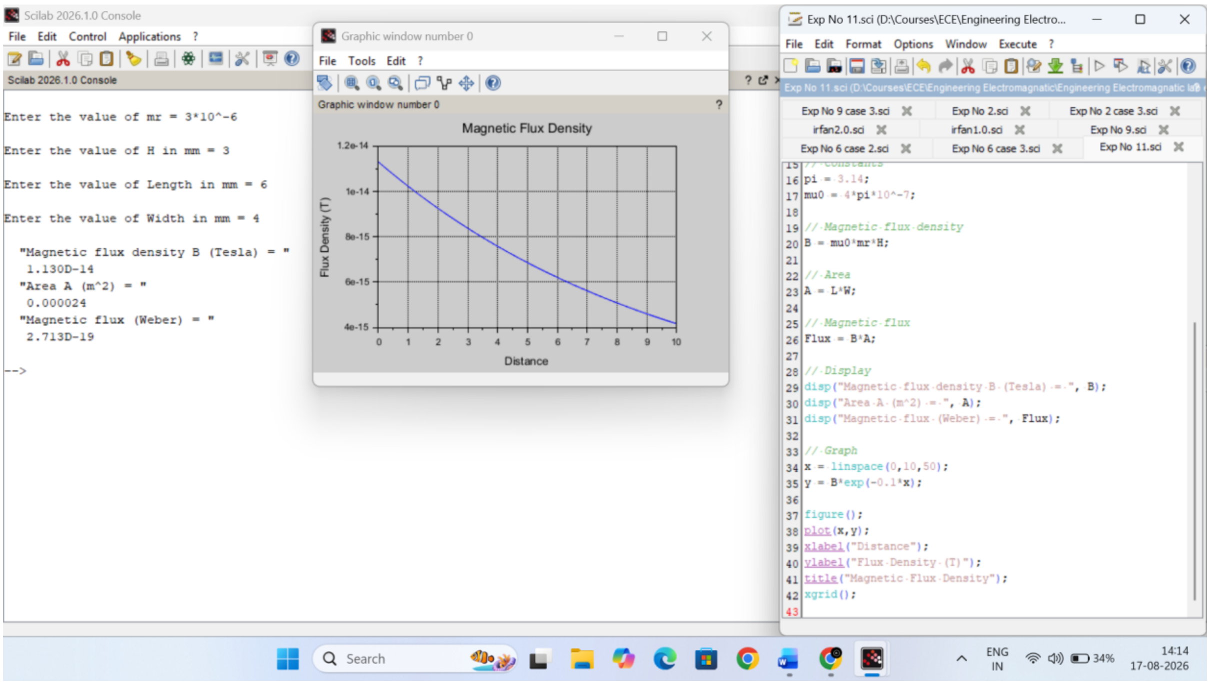
Determination and plotting of Magnetic Flux density in Ferromagnetic Materials with different (i) length, (ii) width, and (iii) magnetic field intensity.

Aim:

To write program for Magnetic flux density in ferromagnetic materials using SCILAB.

Software required:

•SCILAB version 6.0.1



Exp No:- 11 Determine magnetic flux density in ferromagnetic material

$mr = 3 \times 10^{-6}$

$H = 3\ mm$

$L = 6\ mm$

$w = 4\ mm$

$\mu_0 = 4\pi \times 10^{-7}$

magnetic flux Density

$B = \mu_0 \mu_r H$

$B = (4 \times 3.14 \times 10^{-7})(3 \times 10^{-6})(3 \times 10^{-3})$

$\boxed{B = 1.1304 \times 10^{-14}\ T}$

Area

$A = L \times W$

$= (6 \times 10^{-3})(4 \times 10^{-3})$

$= 24 \times 10^{-6}$

$\boxed{A = 2.4 \times 10^{-5}\ m^2}$

$\boxed{0.00024}$

magnetic flux

$\phi = B \times A$

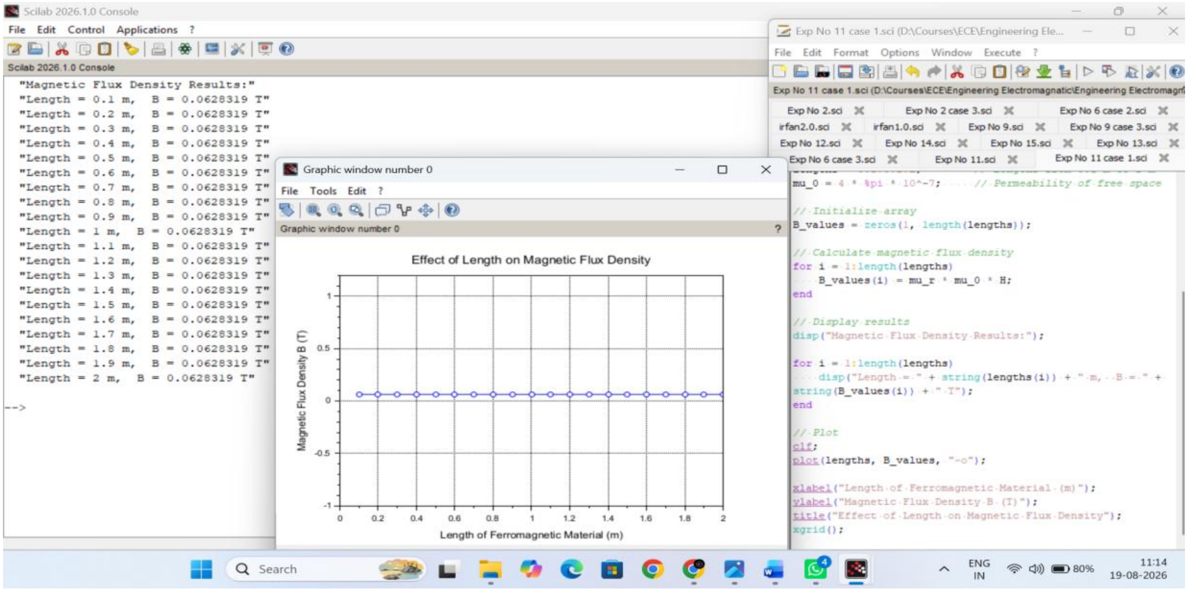
$= (1.1304 \times 10^{-14})(2.4 \times 10^{-5})$

$\boxed{\phi = 2.71296 \times 10^{-19}\ wb}$

Reult :

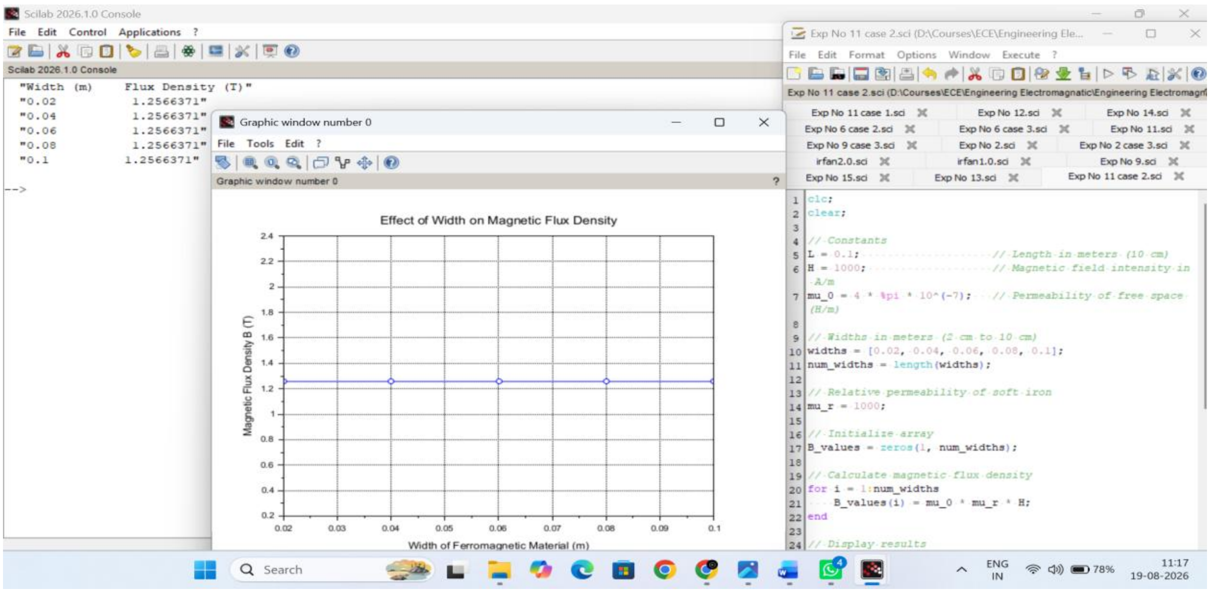
Thus the program was verified for program for Magnetic flux density in ferromagnetic materials using SCILAB was successfully.

Case 1: Examine the effect of length of a ferromagnetic material on the magnetic flux density (B).



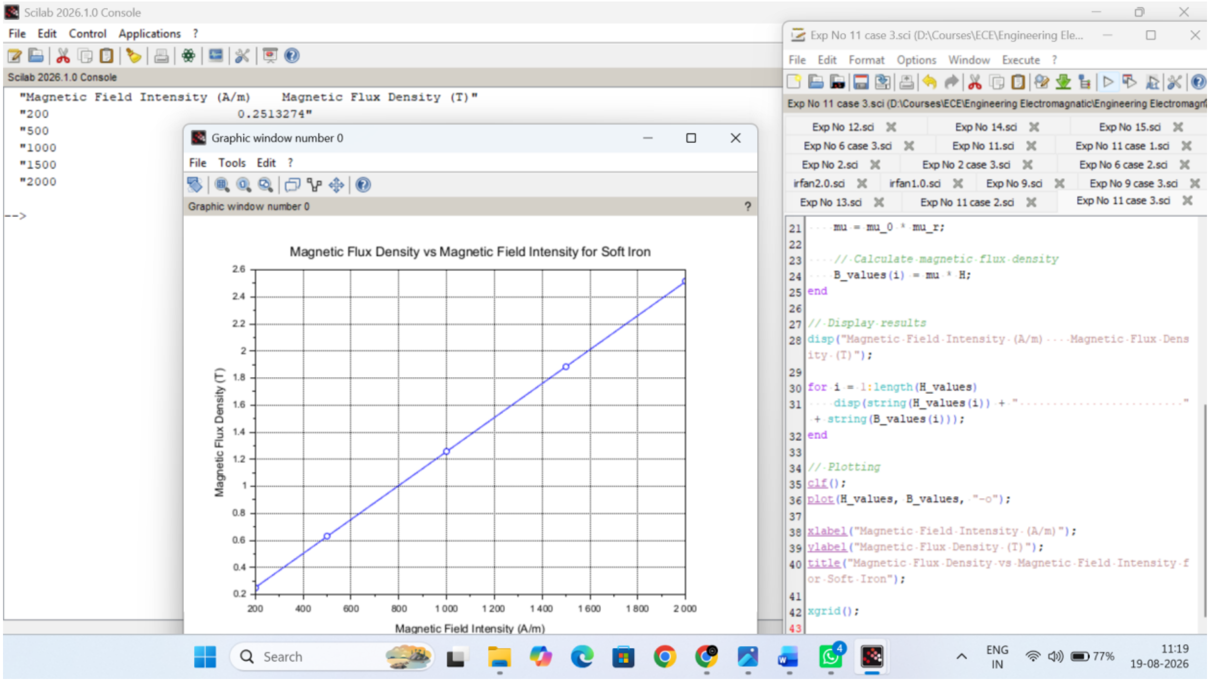
Result: The effect of **length** of a ferromagnetic material on the magnetic flux density (B) is examined.

Case 2: To investigate how the magnetic flux density (B) in a ferromagnetic material change with different widths at a constant length and magnetic field intensity.



Result: The effect of **width** of a ferromagnetic material on the magnetic flux density (B) is examined.

Case 3: To examine how the magnetic flux density (B) in a ferromagnetic material change with different values of magnetic field intensity at constant length and width.



Result:

The effect of Magnetic field intensity of a ferromagnetic material on the magnetic flux density (B) is examined.